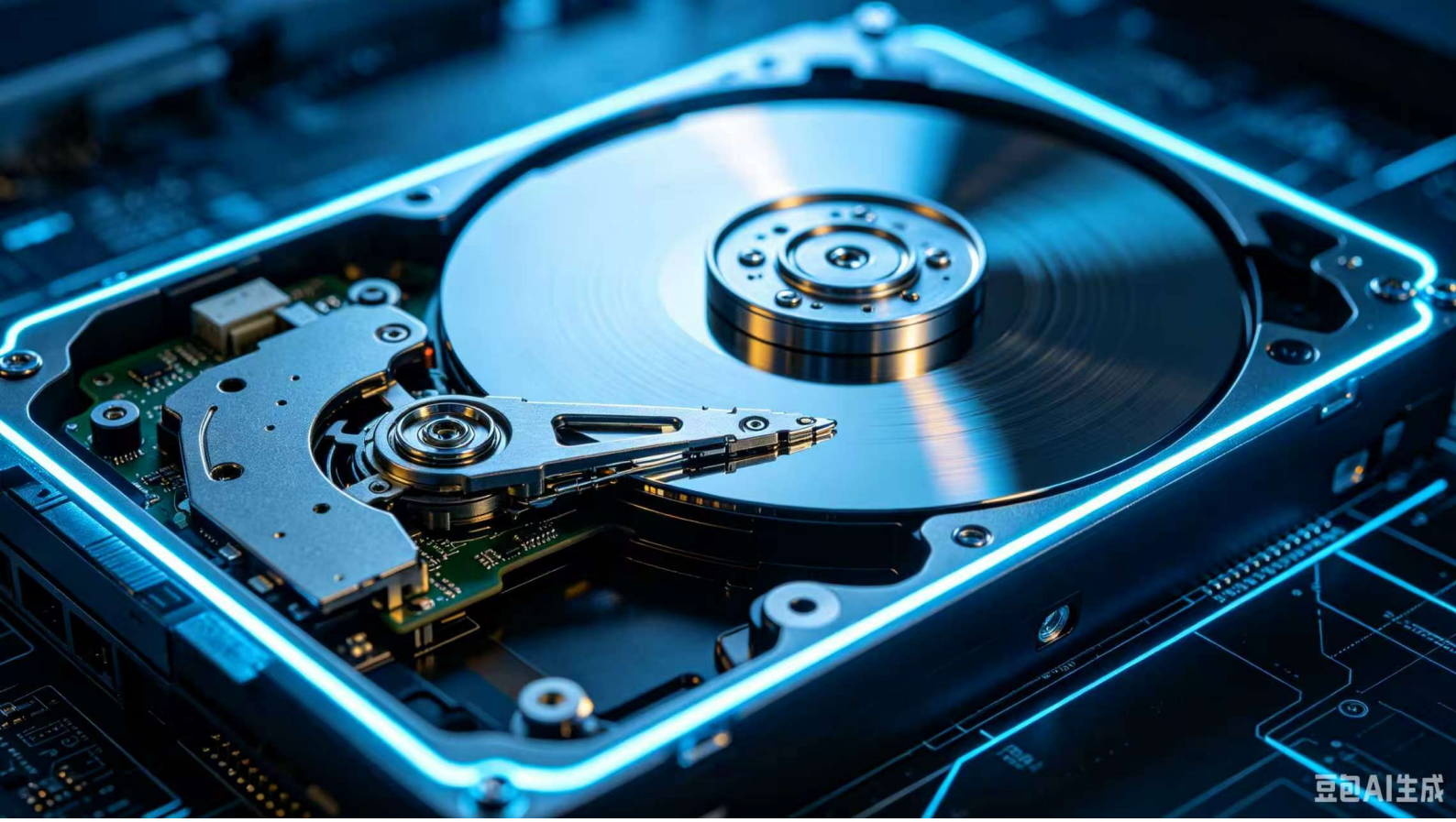


13. Alternating-Current Circuits

> We cannot solve problems with the same thinking we used to create them.
> – *attributed to Albert Einstein*

Since we have introduced the inductance in the last chapter, we now have three elemental electromagnetic devices — resistor, capacitor, and inductor. In many situations, our complicated circuits can be viewed as combinations of these elemental devices.

In earlier studies of RL and RC circuits, we observed that physical quantities such as charge, current, and potential difference either grow or decay exponentially over time. The characteristic time scale for these processes is determined by time constants, which for capacitive and inductive circuits are given by

$$\tau_C = RC, \qquad \tau_L = \frac{L}{R}$$

where R is resistance, C is capacitance, and L is inductance. Even without detailed calculation, we immediately see that a characteristic time constant $\tau_{LC} = \sqrt{LC}$ can be introduced for an LC circuit, which consists of only an inductor and a capacitor connected together. What kind of property does this constant reflect?

Three fundamental time scales. Note what sets τ_{LC} apart from the other two. Both τ_C and τ_L are *decay* times: a capacitor discharges, and the current through an inductor dies away, on those scales,

and after a few multiples of either nothing is left. An LC circuit has no resistance and therefore dissipates nothing, so its time scale sets the *period* of an oscillation, $T = 2\pi\tau_{LC} = 2\pi\sqrt{LC}$, rather than a decay time. The charge swings back and forth indefinitely at the angular frequency $\omega_0 = 1/\tau_{LC} = 1/\sqrt{LC}$, and the amplitude never changes.

13.1 Nature of LC Oscillations

Unlike RL and RC circuits, where exponential growth or decay dominates, an LC circuit exhibits periodic, sinusoidal variations in charge, current, and potential difference. These variations are known as **electromagnetic oscillations**, as energy oscillates between the electric field of the capacitor and the magnetic field of the inductor. In the absence of resistance, these oscillations continue indefinitely, with energy being conserved and merely transferred back and forth between the capacitor and the inductor.

Let's assume that at an arbitrary time t, the current in this LC circuit is given by i, whereas the charge on the capacitor is given by q. Since the circuit has no resistance and no source of emf, the total energy conservation then requires:

$$U = U_B + U_E = \frac{Li^2}{2} + \frac{q^2}{2C}$$

is constant. This yields:

$$\frac{dU}{dt} = \frac{d}{dt}\left(\frac{Li^2}{2} + \frac{q^2}{2C}\right) = Li\frac{di}{dt} + \frac{q}{C}\frac{dq}{dt} = 0$$

Substituting $i = dq/dt$ and $di/dt = d^2q/dt^2$, we obtain the differential equation that governs the charge:

$$L\frac{d^2q}{dt^2} + \frac{1}{C}q = 0$$

This is exactly the standard form of the simple harmonic oscillator equation. For comparison, the mechanical oscillator (such as a block attached to a spring) obeys

$$m\frac{d^2x}{dt^2} + kx = 0$$

with the general solution

$$x = A\cos(\omega_0 t + \phi)$$

where $\omega_0 = \sqrt{k/m}$. Therefore by making the following analogue:

- Charge q corresponds to displacement x,
- $1/C$ corresponds to the spring constant k,
- Current i corresponds to velocity $v = dx/dt$,
- Inductance L corresponds to mass m.

The angular frequency of oscillation for the ideal (resistanceless) LC circuit is then

$$\omega_0 = \frac{1}{\tau_{LC}} = \frac{1}{\sqrt{LC}} \tag{13.1}$$

which directly parallels the angular frequency of a mechanical oscillator, $\omega_0 = \sqrt{k/m}$.

By analogy, the general solution for the charge in the LC circuit is

$$q = Q\cos(\omega_0 t + \phi)$$

where $\omega_0 = 1/\sqrt{LC}$, Q is the amplitude (maximum charge), and ϕ is the phase constant. These constants are determined by the initial conditions:

$$Q\cos\phi = q\big|_{t=0},$$
$$-\omega_0 Q\sin\phi = \left.\frac{dq}{dt}\right|_{t=0} = i\big|_{t=0}.$$

Taking the time derivative of q gives the current:

$$i = \frac{dq}{dt} = -Q\omega_0\sin(\omega_0 t + \phi)$$

Defining $I = \omega_0 Q$, we can also write

$$i = I\cos(\omega_0 t + \phi')$$

where $\phi' = \phi + \frac{\pi}{2}$. Thus, the current leads the charge (and the voltage across the capacitor) by a phase of $\pi/2$.

With above solution, we can also see:

- The maximum values of U_E and U_B are both $Q^2/2C$.
- At any instant, the sum $U_E + U_B$ is always equal to $Q^2/2C$, a constant.
- When U_E is at its maximum, U_B is zero, and vice versa.

In summary, the LC circuit provides a clear example of undamped electromagnetic oscillations, with energy continuously oscillating between electric and magnetic forms, analogous to the oscillation of energy between kinetic and potential forms in a mechanical simple harmonic oscillator.

13.2 Damped Oscillations in Real LC Circuits

In a real LC circuit, the oscillations do not continue indefinitely. This is because real LC circuits lose energy as they oscillate. Here we consider resistive losses, represented by a resistance R, in which the energy of the fields is turned into heat.

This process is analogous to the decay of mechanical oscillations in a block-spring system caused by frictional damping. Just as friction causes a mechanical oscillator to lose energy and eventually come to rest, resistance in an electrical circuit causes the amplitude of oscillations to decrease with time. With a resistance R present, the total electromagnetic energy U of the circuit is no longer constant. Instead, it decreases with time as energy is converted to heat in the resistor:

$$\frac{dU}{dt} = -i^2 R$$

Because of this energy loss, the oscillations of charge, current, and potential difference decrease in amplitude and are said to be **damped**, similar to a damped mechanical oscillator.

The differential equation for the damped RLC circuit is

$$\frac{dU}{dt} = Li\frac{di}{dt} + \frac{q}{C}\frac{dq}{dt} = -i^2 R$$

With $i = dq/dt$ and $di/dt = d^2q/dt^2$, this leads to

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = 0$$

In order to solve this equation, we introduce the *complex formalism*. Previously, in an *RC* circuit, the charge discharges as $q = Qe^{-t/\tau_C}$; in an ideal *LC* circuit, by contrast, the solution is $q = Q\cos(\omega_0 t + \phi) = \Re\left[Qe^{i\omega_0 t}e^{i\phi}\right]$. We notice that both solutions take the form of a constant times an exponential function of time. Therefore, for a general damped oscillation, we assume a solution of the form:

$$q = \Re\left[\tilde{Q}e^{i\tilde{\omega}t}\right],$$

which, when ω is real, is $Qe^{-t/\tau_d}\cos(\omega t + \phi)$. Here $\tilde{Q} = Qe^{i\phi}$ is a complex amplitude and $\tilde{\omega} = \omega + i/\tau_d$ is a complex frequency, with $\tau_d = 2L/R = 2\tau_L$ the **damping time**, on which the amplitude of the oscillation decays.

If we substitute $q = \tilde{Q}e^{i\tilde{\omega}t}$, we obtain

$$-L\tilde{\omega}^2 + i\tilde{\omega}R + \frac{1}{C} = 0$$

This is a quadratic equation in the complex frequency $\tilde{\omega}$, which we solve with the quadratic formula. This gives

$$\tilde{\omega} = i\frac{R}{2L} \pm \sqrt{\omega_0^2 - (1/\tau_d)^2} = \frac{i}{\tau_d} \pm \omega \tag{13.2}$$

where $\omega = \sqrt{\omega_0^2 - (1/\tau_d)^2}$ and $\omega_0 = 1/\sqrt{LC}$. Here, ω_0 is the natural angular frequency of the undamped circuit.

Thus, the behavior of the circuit depends on the relative sizes of $1/\tau_d$ and ω_0:

- **Underdamped:** If $1/\tau_d < \omega_0$, then ω is real and the system oscillates with decreasing amplitude. The circuit eventually comes to rest at equilibrium as energy is dissipated. The solution in this case can therefore be written as:

  $$q = Qe^{-t/\tau_d}\cos(\omega t + \phi)$$

- **Overdamped:** If $1/\tau_d > \omega_0$, ω becomes purely imaginary, and the system does not oscillate; instead, it returns to equilibrium without oscillation. The solution in this case is

  $$q = Q_1 e^{-t/\tau_1} + Q_2 e^{-t/\tau_2}, \qquad \frac{1}{\tau_{1,2}} = \frac{R}{2L} \pm \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}},$$

  a sum of two decaying exponentials, with no oscillation.

- **Critically damped:** If $1/\tau_d = \omega_0$, the two roots coincide at $\tilde{\omega} = i/\tau_d$, so that $e^{i\tilde{\omega}t} = e^{-t/\tau_d}$, and the system returns to equilibrium as quickly as possible without oscillating. Critical damping is often desirable in engineering design for rapid stabilization. The general solution in this case is

  $$q = (Q_1 + Q_2 t)\, e^{-t/\tau_d},$$

  where the two constants are again fixed by the initial charge and current. The factor of t is what a single exponential cannot supply: the two roots of the overdamped case have merged into one, so the second independent solution is t times the first.

13.3 AC Circuits and Forced Oscillations

Oscillations in an RLC circuit will not die out if an external source supplies enough energy to compensate for the energy lost as heat in the resistor. Such a source provides an **alternating current** (ac), in which the emf and current oscillate sinusoidally with time.

The AC generator. An alternating emf is typically produced by an AC generator, which consists of a coil of N turns, each of area A, rotating at constant angular speed about an axis that lies in the plane of the turns and is perpendicular to a uniform magnetic field. Slip rings and metal brushes maintain electrical contact with the rotating coil. The flux through the coil changes as it turns, and the motional emf (Chapter 11chapter.11) is

$$\mathcal{E} = \mathcal{E}_m \cos \omega_d t,$$

where ω_d is the angular frequency of rotation (the **driving angular frequency**) and $\mathcal{E}_m = NBA\omega_d$ is the maximum emf.

The current oscillates at the same frequency and exhibits sinusoidal behavior after a sufficiently long time. We are not interested in the transient behavior here. That transient obeys the undriven equation of Sec. 13.2section.13.2 and dies away with it: on the damping time $\tau_d = 2L/R$ when the circuit is underdamped, and more slowly when it is overdamped. In either case it suffices to describe the steady state. When an external alternating emf is applied, the oscillations of charge, potential difference, and current can be viewed as *driven* by the external emf. In this case, the oscillations occur at the **driving angular frequency** ω_d of the external source, instead of the circuit's natural frequency $\omega_0 = 1/\sqrt{LC}$.

In this case, the energy is supplied by the oscillating emf and current. When the generator is connected to a circuit, it drives a current of the form

$$i = I \cos(\omega_d t - \phi)$$

where I is the amplitude of the current and ϕ is the angle by which the current lags the emf.

Before we proceed, let's look at three simple AC circuits.

Consider a circuit element (resistor, capacitor, or inductor) connected to an alternating voltage source. Take $v(t)$ as the potential of the terminal at which the current enters minus that of the terminal at which it leaves. Then $v = Ri$ for a resistor, $v = L\,di/dt$ for an ideal inductor (Chapter 12chapter.12), and $q = Cv$ with $i = dq/dt$ for a capacitor. The potential difference across the element can be written as

$$v(t) = \Re\left(\tilde{V} e^{i\omega_d t}\right)$$

and the current through the element as

$$i(t) = \Re\left(\tilde{I} e^{i\omega_d t}\right)$$

where $\tilde{V}$ and $\tilde{I}$ are complex amplitudes, and ω_d is the driving angular frequency. Replacing each d/dt in the three relations above by a factor $i\omega_d$, they give $\tilde{Z} = R$, $i\omega_d L$ and $1/(i\omega_d C)$ for the resistor, the inductor and the capacitor in turn.

In AC circuit analysis, it is convenient to define the **complex impedance**:

$$\tilde{Z} = Ze^{i\phi} = \frac{\tilde{V}}{\tilde{I}}$$

where Z is the magnitude of the impedance and ϕ is the angle by which the current lags the voltage.

13.3.1 Resistor

For a purely resistive load,

$$\tilde{Z} = R$$

In this case, voltage and current are in phase.

13.3.2 Inductor

For a purely inductive load,

$$\tilde{Z} = i\omega_d L$$

where L is the inductance. The current lags the voltage by a phase of $\pi/2$.

13.3.3 Capacitor

For a purely capacitive load,

$$\tilde{Z} = \frac{1}{i\omega_d C}$$

where C is the capacitance. The current leads the voltage by a phase of $\pi/2$.

Now, when a resistor R, an inductor L, and a capacitor C are connected in series, the total complex impedance is the sum of the individual impedances:

$$\tilde{Z} = R + i\left(\omega_d L - \frac{1}{\omega_d C}\right) \tag{13.3}$$

It is often useful to express the impedance in polar form:

$$\begin{aligned} \tilde{Z} &= Ze^{i\phi}, \\ Z\cos\phi &= R, \\ Z\sin\phi &= \omega_d L - \frac{1}{\omega_d C}. \end{aligned}$$

From these, we find

$$\begin{aligned} Z &= \sqrt{R^2 + \left[\omega_d L - \frac{1}{\omega_d C}\right]^2}, \\ \tan\phi &= \frac{\omega_d L - \dfrac{1}{\omega_d C}}{R}. \end{aligned}$$

Here, Z is the total impedance of the series circuit, and ϕ is the phase angle by which the current lags the emf.

Impedances in series add, $\tilde{Z} = \tilde{Z}_1 + \tilde{Z}_2$; impedances in parallel add as reciprocals, $1/\tilde{Z} = 1/\tilde{Z}_1 + 1/\tilde{Z}_2$—exactly as resistors do. Elements in series carry the same current and their voltages add, which is the loop rule; elements in parallel have the same voltage and their currents add, which is the node rule. With v and i taken as above, the two rules hold of the complex amplitudes unchanged. A parallel LC pair therefore has

$$\frac{1}{\tilde{Z}} = \frac{1}{i\omega_d L} + i\omega_d C = i\left(\omega_d C - \frac{1}{\omega_d L}\right) \quad \Longrightarrow \quad \tilde{Z} = \frac{i\omega_d L}{1 - \omega_d^2 LC},$$

which diverges at $\omega_d = \omega_0$: the dual of the series resonance below, where the impedance instead falls to its minimum R.

One word on how ac voltages are quoted. The amplitudes $\mathcal{E}_m$, $|\tilde{V}|$ and I used in this chapter are *peak* values. Mains voltages and most ac ratings are instead quoted as **root-mean-square** (rms) values, defined by $V_{\text{rms}} = \sqrt{\langle v^2 \rangle} = |\tilde{V}|/\sqrt{2}$, so that an rms voltage delivers the same average power to a resistor as a dc voltage of the same size. In this book every quoted ac voltage is a peak amplitude unless it is explicitly marked rms.

When the driving angular frequency ω_d matches the natural angular frequency ω_0 of the circuit—that is, at resonance—several things happen at once:

$$\omega_d = \omega_0 = \frac{1}{\sqrt{LC}}. \tag{13.4}$$

- The inductive and capacitive reactances are equal in magnitude, so $\omega_0 L = 1/(\omega_0 C)$.
- The impedance Z is minimized and equals R.
- The current amplitude reaches its maximum value, $I = \mathcal{E}_m/R$, where $\mathcal{E}_m$ is the maximum emf.
- The current and emf are in phase ($\phi = 0$).

We call this situation *resonance*. At resonance, the circuit responds most strongly to the driving emf, a principle that underlies many applications in electronics and communications.

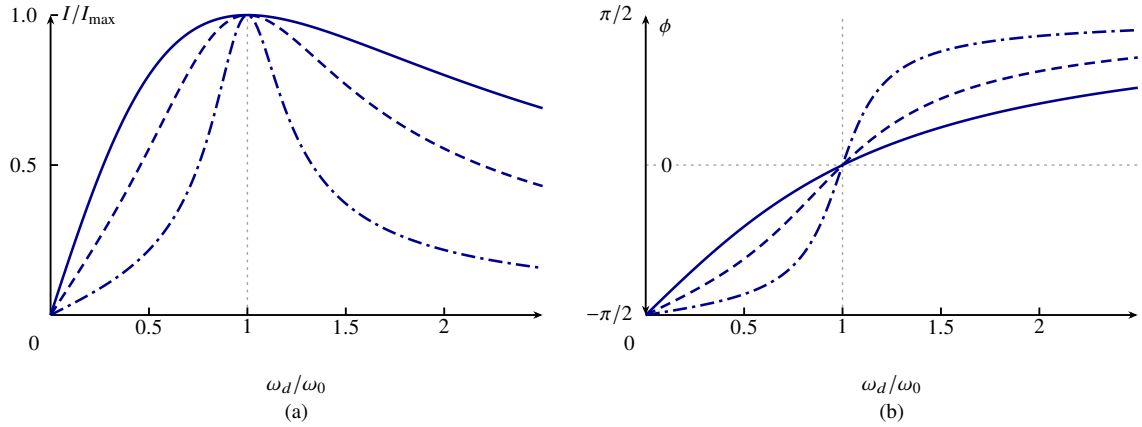


Figure 13.1: The resonance curve. (a) The normalized current amplitude I/I_{max} and (b) the angle ϕ by which the current lags the emf, as functions of the driving frequency ω_d, for a series RLC circuit with L and C held fixed and three values of R: solid, $R = 2\sqrt{L/C}$; dashed, $R = \sqrt{L/C}$; dash-dotted, $R = \frac{1}{3}\sqrt{L/C}$. In (a) all three curves peak at $\omega_d = \omega_0$, where $Z = R$ and the current is in phase with the emf; in (b) all three cross zero there. The smaller R is, the sharper the peak. Below resonance the current leads the emf ($\phi < 0$); above it the current lags ($\phi > 0$).

Resonance curve. With L and C fixed, as in Figure 13.1figure.13.1, the width of the peak is set by R alone: a small R gives a taller, narrower peak of the un-normalized current, a large R a lower and broader one. (Panel (a) plots the current normalized to I/I_{max}, so there all three peaks reach the same height and only their widths differ.) That sensitivity to ω_d is what makes a series RLC circuit a *tuner*—at resonance it responds strongly to one frequency and only weakly to frequencies several peak widths away, and the sharper the peak, the more selective the tuning.

13.4 Exercises

1. Consider a series RL circuit with $R = 300\ \Omega$ and $L = 0.9$ H, driven in the steady state by an emf of fixed magnitude 50 V and fixed angular frequency of 1000 rad/s.

   (a) Calculate the impedance Z of the circuit.

   (b) Calculate the amplitude of the current.

   (c) Calculate the angle ϕ by which the current lags the emf.

2. In the steady state, calculate the amplitude of the current through the resistor in the RLC circuit of Figure 13.2figure.13.2.

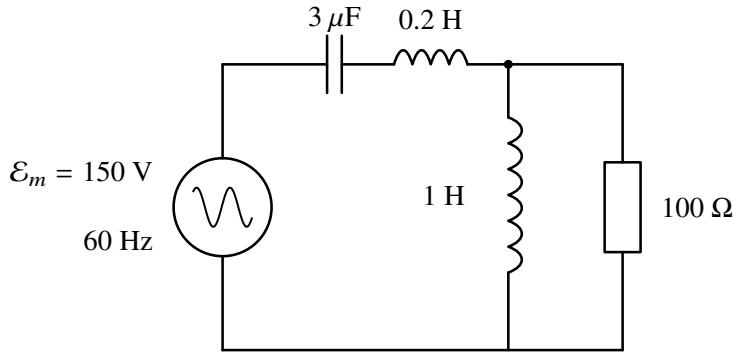


Figure 13.2: RLC circuit for Exercise 2.

3. Consider the circuit of Figure 13.3figure.13.3 in the steady state; the output terminals are left open, so no current is drawn from U_2, and the inductors are ideal, with no resistance. If the amplitude of the output voltage U_2 is to be $1/10$ of the amplitude of the input voltage U_1, then what is the value of the self-inductance L?

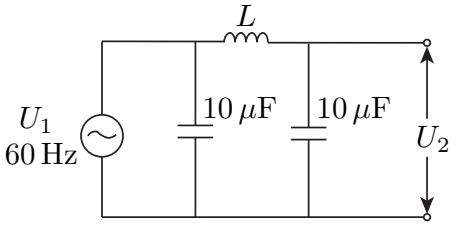


Figure 13.3: Circuit for Exercise 3.

4. Two identical RLC circuits, each with inductance L, capacitance C and resistance R, are coupled through a mutual inductance M (Section 12.5section.12.5). Assume $|M| < L$, and count each charge and current positive in the senses for which $M > 0$.

   (a) Assume that the capacitor in one of the circuits carries charge q_1 and the capacitor in the other circuit carries charge q_2. Write down the coupled differential equations for charges q_1 and q_2.

   (b) Introduce normal coordinates $q_+ \equiv q_1 + q_2$ and $q_- \equiv q_1 - q_2$, show that the differential equations decouple into two independent damped oscillators.

(c) Setting $R = 0$, show that the normal mode frequencies are

$$\omega_+ = \frac{1}{\sqrt{(L+M)C}}, \quad \omega_- = \frac{1}{\sqrt{(L-M)C}},$$

so that the symmetric mode q_+ is the slower of the two, its effective inductance being $L + M$.

5. An ideal battery of emf $\mathcal{E} > 0$, a resistor $r > 0$ and a switch are connected in series between two points A and B, with the battery's positive terminal on the side of A. An ideal inductor L (no resistance) is also connected between A and B, and so is a capacitor C. The switch has been closed for a long time, and it is opened at $t = 0$; after that the inductor and the capacitor form a closed LC circuit. Let i_L be the current in the inductor, counted positive from A to B through the inductor, and q the charge on the plate of the capacitor that is joined to A.

   (a) Just before $t = 0$, find i_L, q and $V_A - V_B$.

   (b) For $t > 0$, find $q(t)$, $i_L(t)$ and $V_A - V_B$. Write $q(t)$ in the form $q = Q\cos(\omega_0 t + \phi)$ of Section 13.1, with $Q > 0$, and give Q, ω_0 and ϕ.

   (c) Find the largest value of $|V_A - V_B|$ for $t > 0$, the first time at which it is reached, and which of A and B is then at the higher potential. Evaluate the voltage, and its ratio to $\mathcal{E}$, for $\mathcal{E} = 12$ V, $r = 6.0\ \Omega$, $L = 0.20$ H and $C = 0.50\ \mu$F, and check it by energy conservation.

6. A capacitor C carries a charge $Q_0 > 0$ on one plate and $-Q_0$ on the other. At $t = 0$ a switch connects it in series with a resistor R and an ideal inductor L, so that the three form a single loop with no source; the current is zero just after $t = 0$. Let $q(t)$ be the charge on the plate that carried $+Q_0$, and $\omega_0 = 1/\sqrt{LC}$.

   (a) Take $R > 2\sqrt{L/C}$. Show that $q(t) = Q_0\,(a\,e^{-bt} - b\,e^{-at})/(a-b)$, and find the decay rates $a > b > 0$.

   (b) Find the leading forms of $1/a$ and $1/b$ when $R \gg 2\sqrt{L/C}$, and compare $1/b$ with the damping time $\tau_d = 2L/R$ of Section 13.2. Which time sets how long the capacitor takes to discharge?

   (c) Keep L and C fixed and let R vary. For $R < 2\sqrt{L/C}$ the charge decays with the envelope e^{-t/τ_d}, at the rate $1/\tau_d$; for $R > 2\sqrt{L/C}$ the slowest rate in $q(t)$ is b. Show that this slowest rate is largest at $R = 2\sqrt{L/C}$, where it equals ω_0.

7. The winding of an AC generator is a flat rectangular coil of N turns, each of area A. It is turned at a constant angular speed ω about the y axis, which lies in its plane and passes through its center, in a uniform, static magnetic field $\mathbf{B} = B\hat{z}$ with $B > 0$. At time t its unit normal is $\hat{n} = \hat{x}\cos\omega t - \hat{z}\sin\omega t$, and a current in the winding is counted positive when it circulates counterclockwise as seen from the tip of $\hat{n}$. Slip rings join the ends of the winding to terminals P and Q, so that a positive current leaves the winding at P. A resistor R and an ideal inductor L are connected in series between P and Q, so that $V_P - V_Q = Ri + L\,di/dt$, where i is the current in the winding. Neglect the resistance and the self-inductance of the winding, friction, and any magnetic coupling between the winding and the inductor L, and consider the steady state. Write $K = NAB$; R, L and ω are positive.

   (a) Find the generator's emf $\mathcal{E}(t)$, counted positive in the positive sense of the winding, and the amplitude I of the current and the angle δ by which the current lags the emf.

(b) Find the y component of the torque that the magnetic field exerts on the winding, and its average over one turn. What torque must the agent turning the winding supply, on average?

(c) For fixed K, R and L, find the angular speed at which the average torque that the agent supplies is largest, and its largest value. How does that average torque behave for $\omega \ll R/L$ and for $\omega \gg R/L$?

8. A coil, which can be treated as an ideal inductor L in series with a resistor $r > 0$, is connected in parallel with a capacitor C. An ideal source of emf $\mathcal{E}_m \cos \omega_d t$ is connected across the pair, and the circuit has reached its steady state. The emf is the potential of the source's terminal A minus that of its terminal B, and the current drawn from the source is counted positive when it leaves the source at A and enters the pair. Write the emf as $\Re(\mathcal{E}_m e^{i\omega_d t})$ and this current as $\Re(\tilde{I} e^{i\omega_d t})$; the impedance of the pair is $\tilde{Z} = \mathcal{E}_m/\tilde{I}$. Let $\omega_0 = 1/\sqrt{LC}$.

(a) Find the complex impedance $\tilde{Z}$ of the pair.

(b) Find the driving angular frequency $\omega_p > 0$ at which the current drawn from the source is in phase with the emf. Show that it exists only if $r < \sqrt{L/C}$, and that $\omega_p < \omega_0$.

(c) At $\omega_d = \omega_p$, find the impedance of the pair, and the ratio of the current amplitude in the coil to the current amplitude drawn from the source. What do both become as $r \to 0$? Compare with the parallel LC pair of Section 13.3.

9. A resistor $R = 10.0\ \Omega$, an ideal inductor $L = 0.100$ H and a capacitor $C = 1.00\ \mu$F are connected in series to an ideal ac source whose emf is 10.0 V rms. The source's angular frequency is set to the resonant value $\omega_0 = 1/\sqrt{LC}$, and the circuit has reached its steady state. Count the current i positive in one sense around the loop, and the emf positive in the same sense. Let q be the charge on the capacitor plate into which a positive i flows, and let v_R, v_L and v_C be the potential drops across the resistor, the inductor and the capacitor, each taken in the sense of i, so that $v_L = L\,di/dt$ and $v_C = q/C$.

(a) Find the peak current and the peak values of v_R, v_L and v_C, first in terms of R, L, C and the rms emf $\mathcal{E}_{\text{rms}}$, then numerically. Compare them with the peak value of the emf.

(b) Show that $v_L + v_C = 0$ at every instant, so that v_R alone equals the emf.

(c) Show that the energy $\frac{1}{2}Li^2 + q^2/2C$ stored in the inductor and the capacitor does not change in time, and find it. Compare it with the heat produced in the resistor in one period.